Projecting Cash Flows

FCFE & FCFF, normalising earnings, leases, R&D, tax rate, capex, working capital and the firm lifecycle

FCFE / FCFF	Normalise	Leases	R&D	Tax rate	Capex & WC	Lifecycle

1. Free Cash Flows

Free cash flow to equity (FCFE)

Net income
↓
plus Depreciation & amortisation
↓
minus Capex & acquisitions
↓
minus Change in non-cash W.C.
↓
plus New borrowings − Debt repaid
↓
= FCFE (potential dividend)

Operating income (EBIT)
↓
minus Taxes on operating income
↓
plus Depreciation & amortisation
↓
minus Capex & acquisitions
↓
minus Change in non-cash W.C.
↓
= FCFF (free cash flow to firm)

REINVESTMENT FOR FUTURE GROWTH

In both flows, the middle items (D&A, capex & acquisitions, change in non-cash W.C.) together make up the ***reinvestment*** *for future growth.*

Three ways to write FCFE

$$FCFE = \text{Net income} - [\,(\text{Capex} - \text{Dep.}) + \Delta\text{Non-cash W.C.}\,] + (\text{New debt issued} - \text{Debt repaid})$$

$$FCFE = \text{Net income} - \text{Reinvestment} + \text{Net debt cash flow}$$

$$FCFE = \text{Net income} \times (1 - \text{Equity reinvestment rate})$$

where **Equity reinvestment rate = (Reinvestment − Net debt CF) / Net income.**

Three ways to write FCFF

$$FCFF = EBIT\,(1 - \text{tax rate}) - [\,(\text{Capex} - \text{Dep.}) + \Delta\text{Non-cash W.C.}\,]$$

$$FCFF = EBIT\,(1 - \text{tax rate}) - \text{Reinvestment}$$

$$FCFF = EBIT\,(1 - \text{tax rate}) \times (\,1 - \text{Reinvestment rate}\,)$$

where **Reinvestment rate = Reinvestment / EBIT (1 − tax rate)**.

2. Estimating Cash Flows

Before using earnings (EBIT or net income) in valuation, always **clean, normalise and update** them. Reported income contains things that do not represent the company's **normal, ongoing earning power**: legal settlements, restructuring expense, R&D expense, operating leases, unusual capex, non-recurring payments etc. Adjust it to clean the numbers.

Normalise earnings

> NOTE
> In this topic (before operating leases) EBIT and net income are used **interchangeably**.

1) Look at the firm's history

Year	EBIT
2023	₹80 cr
2024	₹120 cr
2025	₹40 cr

You might conclude normal EBIT is around **₹80–100 cr**, as ₹40 cr is unusual.

2) Look at comparable companies

Competitor	EBIT margin
Co. A	12%
Co. B	11%
Co. C	13%
Your company	7%

Your company earns 7% while peers earn 11-13%. You might **normalise it towards a sustainable level**.

3. Operating Leases: Convert into Debt

> EXAMPLE
> Annual operating lease payment = **₹20 lakh**. Remaining lease life = **5 years**. Company pre-tax K_d = **10%**.

Step 1: PV of future lease payments

Lease debt = PV (₹20 lakh for 5 yrs @ 10%) = 20 × 3.791 = ₹75.80 lakh

Step 2: split the lease payment

Component	Calculation	Amount
Interest	Lease debt × K_d = 75.80 × 10%	₹7.58 lakh
Principal repayment	₹20 lakh − ₹7.58 lakh	₹12.42 lakh

Step 3: add back lease payment, deduct depreciation on the leased asset

The lease payment was incurred in the P&L, so add it back and deduct depreciation on the leased asset (straight line: 75.80 / 5 = 15.16).

Adjusted EBIT = 100 (reported) + 20 − 15.16 = ₹104.84 lakh

> NOTE
> The depreciation treatment is done **only for US GAAP**.

If net income was taken as the base (for FCFE), also remove the additional interest on the lease:

Adjusted net income = 100 (reported) + 20 − 15.16 − 7.58 = ₹97.26 lakh

Why bother adjusting?

	Company A	Company B
Depreciation	100	–
Interest	20	–
Lease expense	–	120

Without adjustment, B appears to have **lower EBIT** because the entire ₹120 is treated as an operating expense. Economically the two companies are similar, so we convert leases to make them **comparable**.

4. R&D: Convert into an Asset

Suppose a company spends ₹100 cr on R&D. In accounting, R&D goes into the P&L as a whole, but economically it should be **amortised** and then go into the P&L. If the entire amount was deducted, **add it back and then deduct the amortisation expense**.

How do we capitalise R&D?

Step 1: Choose an amortisable life
say 5 years

↓

Step 2: Collect historical R&D spending

↓

Step 3: Find how much of each year's R&D is still unamortised

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Assume the R&D expense is incurred at **year end**. For a 5-year life the unamortised share is:

Year	Current	−1	−2	−3	−4	−5
Unamortised	100%	80%	60%	40%	20%	0%

Example: SAP (year end, € million)

Year	R&D spend	Unamortised %	R&D asset (unamortised)
Current year	1,020.02	100%	1,020.02
−1	993.99	80%	795.19
−2	909.39	60%	545.63
−3	898.25	40%	359.30
−4	969.38	20%	193.88
−5	744.67	0%	0
SAP research asset			**€2.914 billion**

Step 4: amortisation of old R&D this year

$$= 1/5 \times (993.99 + 909.39 + 898.25 + 969.38 + 744.67) \approx €903m$$

Step 5: adjust operating income

$$\text{Adjusted EBIT} = \text{Reported EBIT} + 1{,}020 - 903 = \text{Reported EBIT} + 117$$

KEY IDEA

Cash flow is **unperturbed**, as cash physically moved out anyway. Here we are just **reclassifying the expense**.

NOTE

Growing R&D firms → positive adjustment (current R&D > amortisation of old R&D).

Declining R&D firms → negative adjustment (current R&D < amortisation).

5. Cleansing & Updating Earnings

Cleanse operating items

	Item	What to do
A	Financial expenses	EBIT should **always be before financing expenses**.
B	Capital expenses	**Capitalise and depreciate** it. Do not treat the entire amount as an operating expense.
C	Non-recurring expenses	Adjust them in the earnings (whichever earnings are taken).

Measured earnings = Normalised + Adjusted

Now update the earnings

- **a)** use **TTM** (trailing twelve month) earnings
- **b)** recent quarterly results
- **c)** management disclosures
- **d)** any other reliable source

EXAMPLE

Q3 FY26 + Q4 FY26 + Q1 FY27 + Q2 FY27 = 30 + 35 + 40 + 45 = **₹150 cr**. This is more useful than an older figure.

Parallel treatment for net income (FCFE approach)

Item	What to do
Non-recurring expense	Remove their **after-tax** effect.
Operating leases	Discussed earlier (section 3).
R&D	(1) Add back R&D after tax. (2) Treat R&D like capex. (3) Amortise R&D after tax.

REMEMBER

There can be more adjustments. **Look at the formula**: you know what you want to achieve and what to adjust to make it consistent. Trust your instincts.

6. Reported Numbers vs Valuation Numbers

> **KEY IDEA**
> Financial statements are prepared under accounting rules, but **valuation requires us to think economically**.

1. TTM: trailing earnings

Using the immediately previous 4 quarters matters especially for:

- small companies
- rapidly growing companies
- highly volatile companies
- companies undergoing restructuring, major acquisitions or divestitures

2. Correcting accounting earnings

Already discussed (financial expenses, capital expenses).

3. Operating leases: are they significant enough to matter?

Industry type	Lease expense / Sales	Examples
Large property footprint	4% – 13%	Air transport, trucking, restaurants, telecom wireless, apparel, real estate operations, retail
Negligible leases	0 – 1%	Homebuilding, renewable energy, life insurance, steel, auto & truck

Collateral effects of treating operating leases as debt

	Effect
A	**Cost of capital falls** due to increased debt, as debt is cheaper than equity.
B	**ROIC falls** as invested capital grows substantially.
C	**Net income is not affected**: the interest part of the lease is deducted from EBIT, so the effect remains the same. It is just reclassifying expense.
D	**Balance sheet was understated on both sides.** Add lease debt and lease asset on both sides.

Ratio effects (not that important)

Item	Operating lease	R&D expense
Operating income	↑	↑ growing R&D, ↓ declining R&D
Net income	No change	Changes
Assets	Lease asset ↑	R&D asset ↑
Debt	Lease debt ↑	No change
Equity	No change	↑ (to balance the B/S, as the R&D asset is now on the asset side)
ROIC	↓	↓ (subjective)
ROE	No change	↓ (subjective)
Book debt ratio	↑	No change
Market debt ratio	↑	No change (market already perceives the R&D asset)

> **NOTE**
> Lease obligations are now far more visible on the balance sheet, but we still use **Damodaran's approach** to find lease debt.

4. R&D expenses

In certain industries R&D spend is large: **biotechnology (drugs / pharmaceuticals), internet software, semiconductors, telecom equipment**. If R&D is **20-40% of sales**, we have to consider it as an investment.

7. One-Time Charges & Accounting Malfeasance

Normalisation: one-time charges

Case	Treatment
Non-recurring and unusual / non-periodic	**Add back entirely.**
Recurring, but incurred only every n years	Take the **equivalent annual charge** (Charge / n), not the full amount.

5. Accounting malfeasance

Aggressive firms report higher earnings than conservative ones. ***Look for six warnings:***

A. Unspecified income
Where did the profit actually come from?

B. Asset sale income
Did the company make money from its business or by selling something?

C. Sudden expense changes
Why did expenses suddenly change dramatically?

D. Restatements
Why does the company keep changing previously reported numbers?

E. Profit > cash
Is accounting profit turning into operating cash flow?

F. Tax income vs reported income
Why is taxable profit different from accounting profit?

WHAT TO DO

Investigate the reason *and decide whether it is temporary, persistent or fraud. Then adjust the numbers if it is temporary or fraud, or if the reason is unjustifiable or unknown.*

If earnings are negative or unusually low

Cause	What to do
A. Macro factor *commodity price drop or recession*	*Estimate* ***sustainable / cycle (average) earnings*** *by looking at other years and the whole industry: how much would it earn in normal conditions?*
B. Young company	*Losing money does not mean a bad business model. Estimate the profit margin mature companies in the business earn and* ***target that margin in the long term****.*
C. Structural problems *the business itself is responsible (bad model or weakness)*	*Use the* ***industry average margin as target*** *and move towards it over time, as structural problems are fixed (if fixable).*

8. Tax Rate

	Effective tax rate	Marginal tax rate
Meaning	Taxes paid / Pre-tax income	Rate on the **next ₹1** of taxable income
Tells about	**Historical** data	**Future** income
Why different	Can be **lower** than statutory / marginal due to tax deductions, tax credits, tax losses carried forward, tax-exempt income	Single country: look up the applicable rate (incl. surcharge, cess). Multinational: **weighted average** marginal rate on future operating income

Multinational: weighted average tax rate

Country	Operating income	Tax rate
US	₹300 cr	30%
India	₹600 cr	25%
UK	₹100 cr	25%

WATR = (300 × 30% + 600 × 25% + 100 × 25%) / 1000 = 26.5%

NOTE

Use the **same tax rate** when calculating after-tax cost of debt: $K_d (1 - t)$.

Effective vs marginal (only read, no need to learn formulas)

Marginal rate can UNDERSTATE early-year after-tax income

Suppose a company has tax losses from previous years. If the losses offset current taxable income, it may pay nil tax in the early years, so effective tax rate = 0% and EBIT (1 − t) should equal EBIT. If we use a marginal rate (say 30%) we say EBIT (1 − t) is **less than** EBIT.

Effective rate can OVERSTATE later-year income

Once the company is consistently profitable and its tax losses are exhausted, it eventually pays the marginal rate. If we keep using the effective rate we overstate income:

$$EBIT (1 - t_e) > EBIT (1 - t_m) \text{ as } t_e < t_m$$

SOLUTION

Start with the **current effective tax rate** and **gradually move towards the marginal tax rate**. To time the move, just follow the **net operating loss (NOL) exhaustion**.

Example: following NOL exhaustion (starting NOL = ₹300 cr)

	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6
EBIT (A)	−50	100	150	200	250	250
Interest (B)	−40	−40	−40	−40	−40	−40
EBT (A − B)	−90	60	110	160	210	210
Beginning NOL (C)	300	390	330	220	60	0
NOL used (D)	–	60	110	160	60	0
Ending NOL	390	330	220	60	0	0
Taxable income (EBT − D)	0	0	0	0	150	210
Tax @ 30% (E)	0	0	0	0	45	63
Effective tax rate	0%	0%	0%	0%	21.43%	30%

READING THE TABLE

When the NOL is fully exhausted in year 5, the effective rate is still not 30%, as ₹60 cr of NOL was used to shield EBT. In year 6 **no NOL is used**, so the marginal rate equals the effective rate.

9. Net Capital Expenditure

Net capex = Capital expenditure − Depreciation

	Meaning
Capex	Money spent on long-term assets. Represents **addition of new assets**.
Depreciation	Spreads cost over the useful life of existing assets. Represents **replacement / consumption of existing assets**, so it is subtracted.

Net capex depends on ***how fast the firm is growing****: higher g → more capacity assets → higher capex. A mature company with stable growth only replaces the worn-out part, so:*

Capex ≈ Depreciation → Net capex ≈ 0 (mature companies)

> NOTE
> *This is* ***not always true****. Check with the formula.*

Capital expenditure: adjustments

1. R&D

Adjusted net capex = Net capex + Current year's R&D − Amortisation of R&D (old)

2. Acquisitions: *buying a target company is economically an investment, as the company spends cash to acquire future earning capacity.*

Adjusted net capex = Net capex + R&D − Amortisation of R&D + Acquisitions − Acquisition amortisation

- *If acquisitions occur once or twice, or are infrequent over a long period,* ***normalise*** *them by dividing over the years.*
- *Find acquisitions in* ***Cash flow from investing activities****.*
- *Acquisition depreciation is based on the depreciation of the* ***acquired assets****.*

10. Working Capital Investment

Working capital is ***money tied up in day-to-day operations****: current assets − current liabilities.*

Include	Exclude
Accounts receivable *and* **inventory** *(current assets)* **Accounts payable** *and* **supplier credit** *(current liabilities)*	**Cash**: *not really an operating investment* **Debt**: *not operating working capital*

Non-cash WC = Inventory + Accounts receivable − Accounts payable − Other operating current liabilities

WHY THESE EXCLUSIONS

Cash is excluded because we are trying to find how much **additional cash** is needed to run the business.

Debt is a financing decision and is handled separately through **cost of capital**.

Working capital is treated as an **investment**.

General propositions

1. Do not over-split. Breaking WC into many components is pointless unless you can forecast each one. **Better:** use a relationship with sales, e.g. WC = 10% of sales (easy to forecast in Excel).

Change in NWC	Cash effect
Increase	**Cash outflow**
Decrease	**Cash inflow**

2. Working capital volatility.

Year	Sales	NWC	NWC / Sales
1	₹1,000m	₹100m	10%
2	₹1,100m	₹150m	13.6%
3	₹1,200m	₹105m	8.75%

NWC goes 100 → 150 → 105, which is very volatile. **Do not use YoY change**, as it will be distorted. Instead look at **NWC / Sales** each year and compare it against historical averages, industry averages and competitors. Choose a reasonable long-term % (say 10%). If your company is at 15% now, you might forecast it **gradually moves towards 10%**.

3. NEGATIVE WORKING CAPITAL

Negative WC is not necessarily bad and, ironically, can be **very good**.

Retailer: customers pay at checkout while we have time to pay suppliers.

Amazon: cash comes in quickly but is reinvested into expansion, so the asset is lower than the credit terms from suppliers.

If the company grows → more sales → more purchases from suppliers. **Increase in negative WC → cash inflow; decrease in negative WC → cash outflow.**

11. Free Cash Flow to Equity (FCFE)

The only cash flow to equity is the ***dividend****. But* ***actual dividend ≠ potential dividend****: the company does not distribute all cash available to shareholders, as it is needed to maintain a stable dividend every year, for future needs and acquisitions. So we must calculate the* ***potential dividend****.*

FIRST MISTAKE

Earnings ≠ Potential dividend. *Three problems:*

A. *Earnings are not cash: some items are on accrual basis, not cash basis.*

B. *The company needs to reinvest.*

C. *Paying all earnings prevents the growth of the company.*

WHAT IS POTENTIAL DIVIDEND?

Cash left over after the company makes the investments required for future growth and handles its debt requirements.

For a leveraged company:

$$\text{FCFE} = \text{Net income} - (\text{Capex} - \text{Depreciation}) - \Delta\text{WC} + (\text{New debt issued} - \text{Debt repayment})$$

Also called ***residual cash flow****, because shareholders are the last claimants.*

FCFE from the statement of cash flows

$$\text{FCFE} = \text{CFO} - \text{Capex} - \text{Cash acquisitions} + (\text{Debt issued} - \text{Debt repaid})$$

Estimating FCFE when leverage is stable

Suppose the company maintains a ***stable debt / capital ratio (D/R)****.*

$$\text{FCFE} = \text{Net income} - (\text{Capex} - \text{Dep.})(1 - \text{D/R}) - (\text{Working capital needs})(1 - \text{D/R})$$

WHY MULTIPLY BY (1 - D/R)?

Say the D/R is 30%. Whatever equity has to finance is reduced by that share, i.e. (1 − 30%) = ***70%****.*

NOTE

Use ***book value*** *when analysing historical data for D/R and* ***market value*** *when forecasting.*

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(Debt issued – Debt repayment) is **embedded** in the formula above. Say we have old debt of ₹200 to repay; we issue ₹200 of new debt only (to keep D/E stable). So **Debt issued = Debt repaid → overall effect nil**, which is why it is not shown explicitly.

The formula assumes D/R is constant, i.e. new debt cannot be issued in excess of the amount required to repay old debt.

12. FCFE across the Lifecycle



Stage	What happens	FCFE
1. Start-up	Reinvestment ↑. Cannot pay dividends but needs shareholders to put in money.	–ve
2. Growth (young)	Growing quickly while profits may still be low. Reinvestment needs are still high. Needs external financing.	May still be –ve
3. High growth	Profitable, but reinvestment is still large. The company starts becoming **self-funding** (cash retained and also given to shareholders).	Turns +ve
4. Mature growth	Growth continues but begins slowing. Reinvestment may fall. Can start paying dividends and equity buybacks.	+ve and growing
5. Mature stable	Not growing aggressively anymore. Reinvestment is low. Pays dividends and buys back equity.	+ve and stable
6. Decline	Business is shrinking and needs no reinvestment; may actually sell assets (divestment). May return capital to shareholders as the business liquidates.	+ve and greater than earnings

FCFE over time: Tesla

Year	Stage	Net income	FCFE
2017	Early / high growth	–$1.96 bn	–$4.93 bn
2021	Mature	+$5.52 bn	+$2.33 bn

In the early years, net income and FCFE are **negative** because of huge spending on factories, equipment and expansion. In 2017 FCFE was **more negative** than net income because the company had to invest heavily. By 2021 it was generating substantial cash for shareholders, but **FCFE < net income** as it was still reinvesting.



13. Is FCFE Leverage a Free Lunch?

As we increase debt, FCFE rises too. But does that mean a higher valuation? **Not necessarily.** More debt is perceived as risky by equity investors, so the cost of equity rises (primarily because **beta increases**). K_o also moves in alignment with K_e, so the **discount rate gets higher**, which might lead to a lower valuation.

Case	What dominates	Result
A	Cash flows dominate the discount rate effect: FCFE ↑ more than risk ↑	**Value ↑**
B	Risk effects dominate: K_e ↑ more than FCFE	**Value ↓**
C	Both offset	**Value unchanged**
D (correct answer)	Any one of the above, depending on the company and its current leverage	**It depends**